

DSA 3050A — Business Intelligence & Visualization

Data Dictionary

Complete column-level definition of the raw and cleaned datasets

Document	Data Dictionary
Group	Grp05
Topic	Option 4 — Airline Delay and Operations Dashboard
Dataset	hflights — 227,496 Houston departures, 2011
Source	US DOT Bureau of Transportation Statistics
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1. Purpose and scope

This dictionary defines every column in both datasets used by the project: the raw extract as downloaded (227,496 rows × 21 columns) and the cleaned table produced by the Power Query pipeline (227,496 rows × 32 columns). For each column it gives the name, the data type as set in the model, the business meaning, an example value taken from the data itself, and the count of missing values.

The row count is identical before and after cleaning. No flights were dropped: the cleaning work renamed, decoded, derived and re-typed, but never filtered. This is deliberate — cancelled and diverted flights are the subject of part of the analysis, so removing them would have destroyed the evidence.

2. Raw dataset — Dataset/raw_dataset.csv

227,496 rows × 21 columns. Source: US DOT Bureau of Transportation Statistics, via the R *hflights* package.

Column	Type	Meaning	Example	Nulls
Year	Whole number	Calendar year of the flight. Constant 2011 throughout, so removed during cleaning.	2011	0
Month	Whole number	Month of departure, 1-12.	1	0
DayOfMonth	Whole number	Day of the month, 1-31.	1	0
DayOfWeek	Whole number	Day of week, 1 = Monday through 7 = Sunday.	6	0
DepTime	Decimal	Actual departure time as an HHMM integer (e.g. 1400 = 14:00). Null when the flight did not depart.	1400.0	2,905
ArrTime	Decimal	Actual arrival time as an HHMM integer. Null when the flight did not arrive.	1500.0	3,066
UniqueCarrier	Text	Two-letter IATA carrier code identifying the operating airline.	AA	0
FlightNum	Whole number	Flight number as published by the carrier. Not unique across carriers.	428	0
TailNum	Text	Aircraft registration (tail number) — identifies the physical airframe.	N576AA	795
ActualElapsedTime	Decimal	Total gate-to-gate time in minutes: taxi-out + air time + taxi-in.	60.0	3,622
AirTime	Decimal	Wheels-up to wheels-down time in minutes.	40.0	3,622
ArrDelay	Decimal	Arrival delay in minutes. Negative means the flight arrived early. Null for cancelled and diverted flights.	-10.0	3,622
DepDelay	Decimal	Departure delay in minutes. Negative means an early pushback. Null for cancelled flights.	0.0	2,905
Origin	Text	Origin airport IATA code — IAH or HOU only.	IAH	0
Dest	Text	Destination airport IATA code. 116 distinct values.	DFW	0
Distance	Whole number	Great-circle distance between the airports, in statute miles.	224	0
TaxiIn	Decimal	Minutes between landing and arriving on stand.	7.0	3,066
TaxiOut	Decimal	Minutes between pushback and take-off.	13.0	2,947
Cancelled	Whole number	Flag: 1 if the flight was cancelled, 0 otherwise.	0	0
CancellationCode	Text	Reason code, populated only when Cancelled = 1. A = Carrier, B = Weather, C = National Air System, D = Security.	A	224,523

Column	Type	Meaning	Example	Nulls
Diverted	Whole number	Flag: 1 if the flight was diverted to an airport other than its scheduled destination.	0	0

3. Cleaned dataset — Dataset/cleaned_dataset.csv

227,496 rows × 32 columns. This is the Fact_Flights table in the Power BI model.

Column	Type	Meaning	Example	Nulls
Flight Date	Date	Calendar date of departure, built from Year, Month and DayofMonth. Joins to Dim_Date[Date].	2011-01-01	0
Month	Whole number	Month number, 1-12. Kept for sorting.	1	0
Month Name	Text	Abbreviated month name, Jan-Dec.	Jan	0
Quarter	Text	Calendar quarter, Q1-Q4.	Q1	0
Season	Text	Winter (Dec-Feb), Spring (Mar-May), Summer (Jun-Aug), Autumn (Sep-Nov).	Winter	0
Day	Whole number	Day of the month, 1-31.	1	0
Day of Week	Text	Abbreviated weekday name, Mon-Sun.	Sat	0
Airline Code	Text	Two-letter IATA carrier code, trimmed and upper-cased.	AA	0
Airline	Text	Full airline name, attached by merging Dim_Airline on Airline Code. Joins to Dim_Airline[Airline].	American Airlines	0
Flight Number	Whole number	Carrier flight number.	428	0
Aircraft ID	Text	Aircraft registration. 795 missing values replaced with "Unknown".	N576AA	0
Origin Airport	Text	Origin IATA code — IAH or HOU.	IAH	0
Destination Airport	Text	Destination IATA code.	DFW	0
Route	Text	Origin and destination merged into one key, e.g. IAH-DFW. The drill-through field.	IAH-DFW	0
Distance (Miles)	Whole number	Great-circle distance in statute miles.	224	0
Distance Band	Text	Short Haul (<500), Medium Haul (500-999), Long Haul (1000-1499), Extended Haul (1500+).	Short Haul	0
Departure Time	Text	Actual departure as HH:MM clock text, converted from the HHMM integer. 2400 rolls to 00:00.	14:00	2,905
Departure Hour	Whole number	Hour of departure, 0-23. Drives the time-of-day analysis.	14.0	2,905
Time of Day	Text	Night (00-05), Morning (06-11), Afternoon (12-17), Evening (18-23).	Afternoon	2,905
Arrival Time	Text	Actual arrival as HH:MM clock text.	15:00	3,066
Departure Delay	Whole number	Departure delay in minutes. Null for cancelled flights — deliberately not imputed.	0.0	2,905
Arrival Delay	Whole number	Arrival delay in minutes. Null for cancelled and diverted flights.	-10.0	3,622
Delay Recovered	Whole number	Departure Delay minus Arrival Delay — minutes clawed back in the air. Positive means time was made up.	10.0	3,622
Delay Category	Text	On Time / Early (≤0), Minor (1-14), Moderate (15-59), Severe (60-179), Extreme (180+), Not Completed (null).	On Time / Early	0
Elapsed Time	Whole number	Gate-to-gate minutes.	60.0	3,622

Column	Type	Meaning	Example	Nulls
Air Time	Whole number	Wheels-up to wheels-down minutes.	40.0	3,622
Taxi In	Whole number	Minutes from landing to stand.	7.0	3,066
Taxi Out	Whole number	Minutes from pushback to take-off.	13.0	2,947
Flight Status	Text	Cancelled, Diverted, On Time (Arrival Delay $\leq$ 0) or Delayed. The measure filter field.	On Time	0
Is Cancelled	Text	Yes / No, decoded from the raw 0/1 flag.	No	0
Is Diverted	Text	Yes / No, decoded from the raw 0/1 flag.	No	0
Cancellation Reason	Text	Carrier, Weather, National Air System, Security, or Not Cancelled.	Not Cancelled	0

4. Dimension tables

Dim_Airline — 15 rows, one per operating carrier.

Column	Type	Meaning
Airline Code	Text	Two-letter IATA carrier code. Merge key against the fact table.
Airline	Text	Full carrier name. The relationship key to Fact_Flights[Airline].

Dim_Date — 365 rows, 1 Jan to 31 Dec 2011. Marked as the model date table.

Column	Type	Meaning
Date	Date	Calendar date. Relationship key to Fact_Flights[Flight Date].
Year	Whole number	Calendar year — 2011 throughout.
Month	Whole number	Month number 1-12, used to sort Month Name.
Month Name	Text	Full month name, January-December.
Quarter	Text	Q1-Q4.
Day	Whole number	Day of month, 1-31.
Day of Week	Text	Full weekday name.
Week of Year	Whole number	ISO week number, 1-52.
Is Weekend	Text	Yes for Saturday and Sunday, otherwise No.

Summary_Airline_Performance — 15 rows, one per carrier, built with Group By.

Column	Type	Meaning
Airline	Text	Carrier name. Relationship key to Dim_Airline[Airline].
Total Flights	Whole number	Count of flights operated by that carrier.
Average Arrival Delay	Decimal	Mean arrival delay in minutes across completed flights.
Total Cancellations	Whole number	Count of cancelled flights for that carrier.

5. Code reference

Cancellation codes as published by the BTS:

Code	Meaning	Count in 2011	Notes
A	Carrier	1,202	Within the airline's control — crew, maintenance, aircraft swap.
B	Weather	1,652	Outside anyone's control. Heavily concentrated in February.
C	National Air System	118	Air traffic control, airport capacity, non-extreme weather.
D	Security	1	Security incidents, evacuations, screening failures.
blank	Not cancelled	224,523	Null by design — the flight operated. Mapped to "Not Cancelled".

Airport codes

Code	Airport	Flights	Notes
IAH	George Bush Intercontinental Airport	175,197	Large international hub. Long taxiways — 16.9 min average taxi-out.
HOU	William P. Hobby Airport	52,299	Smaller domestic airport. 9.1 min average taxi-out.

A note on nulls in the delay columns. The 3,622 null values in Arrival Delay and 2,905 in Departure Delay are not data-quality defects. They are the cancelled and diverted flights, which have no arrival or departure to be late for. They were deliberately left null rather than filled with zero: a zero would read as "on time" and would have pulled every delay average towards zero while inflating the on-time rate. All rate measures in the model divide by completed flights instead.